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Published June 16, 2014



# Integrating a Mobile-Based Gaming Application into a Postsecondary Forest Ecology Course

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**ABSTRACT** Increased disengagement of the current generation of postsecondary students (sometimes referred as "net generation") from traditional instruction coupled with on-going popularity of games and mobile technologies have prompted interest in game-based learning in education. The objective of this study was to develop and evaluate the viability of a mobile game-based learning quest, based on the Questogo platform (website and mobile app) in an undergraduate Forest Ecology course offered at the University of British Columbia (UBC), Vancouver. The disturbance ecology (DE) quest was designed as a self-study activity that supports field-based laboratory sections of the course. The quest included instructional, location-based, and question- and answer-type of tasks that tested students' knowledge of forest and disturbance ecology in an outdoor setting. After completing the DE quest, students provided feedback via an online survey. The majority of students found the DE quest to be a useful self-study tool, with 81% of respondents indicating that they were able to successfully engage with the mobile game-based learning technology. Sixty-six percent of the students would like to also see quests incorporated into other courses and 28% would like to have additional quests in the Forest Ecology course. This study provides a framework for incorporating mobile game-based learning into outdoor learning activities that offer students an engaging self-study educational experience.

Use of mobile devices such as mobile or cell phones, namely smartphones, as well as tablets, is growing in popularity across the world due to recent developments in communication and wireless technologies. In 2011 in Canada alone, there was a 50% increase in mobile phone users (Open Door Technology, 2011) and it is estimated that Canadian mobile device users are spending up to 17.3 hours per week for activities ranging from entertainment to education. According to a study by Quitney-Anderson and Rainie (2008), mobile devices will be the primary connection to the Internet for most people in the world by 2020. As mobile devices become ever more embedded in daily life, they have also entered postsecondary educational institutions, prompting educators and researchers to take a pedagogical view toward developing educational applications for mobile devices to promote teaching and learning (Wu et al., 2012).

In the changing postsecondary educational environment it is important to develop approaches that do not just focus on content, but also allow learners to organize and apply that content in problem-solving, real-life situations. Mobile technologies can help students by empowering them to observe and collect data about and within their local natural environments, enhancing their experience of nature and ecological processes. New technologies offer numerous opportunities to engage students in learning about the natural environment, enabling them to develop 21st century literacies such as enhanced informational, digital, and visual skills (Thompson et al., 2003). Mobile devices, through their access to the Internet, cameras, and global positioning system (GPS), make it easy to gather, organize, and submit data from various kinds of observations. Mobile applications (so called "apps") can be downloaded to engage users in citizen science activities, like documenting wildlife (e.g., Project Noah App, Networked Organisms, 2013), or tree pests and diseases (e.g., Tree Alert App, Forestry Commission UK, 2013), or reporting invasive plant species (e.g., Invasive Alien Plant Program, 2013). Currently, there are also mobile apps that allow users to create and play their own games in their own environment via mobile phones, GPS, and the Internet. This offers opportunities for active engagement in learning, which is constructive, contextual, reflective, and social; emphasizing the constructivist approaches to learning with technology as described by Jonassen et al. (1999) and Barak (2006).

The present generation of postsecondary students has grown up with information communication technology and they handle digital information on a daily basis, are connected to each other via mobile technologies, work interactively, often perform several tasks simultaneously, and play games to a greater extent than previous generations. According to Prensky (2001) and van Eck

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Nat. Sci. Educ. 43:117–125 (2014)
Available freely online through the author-supported open access option.
doi:10.4195/nse2014.02.0004

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Abbreviations: DE, Disturbance Ecology; GPS, global positioning system; PSR, Pacific Spirit Regional; UBC, University of British Columbia, Vancouver.

(2006), increased disengagement of the so-called "net generation" or "digital natives" from traditional instruction together with on-going popularity of games have prompted interest in game-based learning at both secondary and postsecondary levels. Although some studies have reported that game-based learning is enhancing students' motivation (Garris et al., 2002), the empirical evidence is far from conclusive that game-based learning is more adequately engaging students and allowing them to better acquire knowledge than traditional learning methods (van Eck and Dempsey, 2002; Schwabe and Göth, 2005; Huizenga et al., 2009).

Mobile and location-based technologies provide opportunities to embed learning in authentic environments and thereby enhance engagement and learning outside traditional educational settings (Klopfer and Squire, 2008). In ecology, a game-based learning that provides an interactive window into the scientific method and includes formulation of research questions, data gathering, reasoning, and conclusion development greatly improves students' understanding of the complex ecological relationships in the natural world (Finn et al., 2002). Given the astounding complexity of natural ecosystems, it is crucial to develop educational tools and approaches that can teach future land-planners and managers how to successfully solve the current global challenges such as climate change and declining species diversity.

The objective of this study was to develop and evaluate the viability of a mobile game-based educational quest. The quest was housed on a mobile smartphone application, including a web version, based on the Questogo platform. The quest was designed as a self-study activity that supports field-based laboratory sections of a lower-level undergraduate Forest Ecology course offered by the Faculty of Forestry at UBC. The quest focused on the topic of disturbance ecology; hereafter referred to as DE quest. The evaluation of the DE quest was conducted during the first year of its implementation in the course mentioned above.

## DEVELOPMENT OF THE DE QUEST

### Questogo Platform

The mobile game-based DE quest described in this article was based on the Questogo platform, developed by the 14Oranges Software Inc. company, Richmond, BC. Questogo is an interactive game, similar to a scavenger hunt, in which users explore new places by navigating through neighborhoods, deciphering clues, conquering challenges, and discovering hidden treasures. User experiences can then be shared on a social platform. Questogo was inspired by Geocaching (Groundspeak, 2014), a popular real-world, outdoor treasure hunting game in which participants use GPS-enabled devices to search for a geocache (container) hidden at a specific set of GPS coordinates. 14Oranges Software Inc. company developed the concept of Questogo as well as its design, website, and free mobile app. Currently, the app provides support for Apple iPhone and iPad only; however, support for the Android operating system, the world's most widely used smartphone platform, is in development.

After signing up to the Questogo platform, users are invited to either play existing scavenger hunts or create their own quests that can in turn be shared openly with other Questogo users or only with a limited audience, such as in our case the students of the undergraduate Forest Ecology course. Questogo also provides a set of analytics, which can be used by course instructors to review users' results, evaluate students, or adapt quests accordingly.

In general, a Questogo-based quest runs off of three distinct types of so-called "tasks" that prompt the users to complete a variety of actions. Users are then rewarded points depending on how well they performed. The points awarded after the correct completion of each task contribute to the user's total score at the end of the quest, and this score can then be shared with peers or with the course instructor only. In a typical task sequence, the first task could be a simple instructional task for which no points are awarded. The second task could be a location-based task, for which participants are prompted to verify their location using the Global Positioning Unit in their Apple platform device, and are awarded points if they are within 15 m of the intended location. A third and final task-version could be a question-and-answer task for which participants need to type in either a short phrase or a single word as an answer. Participants are awarded points if their answer matched one of the possible answers entered previously by the quest creator.

Question and location tasks can include hints to assist the participants in reaching the correct answer or location; however, these hints require the participants to forfeit some of their awarded points to obtain the hints. The three different task-versions can be arranged in any manner to the quest-creators' liking. Questogo also allows creators to write questions with different levels of difficulty and to match question difficulty-levels with a ranking system of points awarded. Thus, questions can be catered to a wide range of users with a varying knowledge base. This allows users to choose between varying levels of complexity and rewards more adept users to answer increasingly challenging and at the same time more point-gaining questions. A limitation of the Questogo platform is, however, that it only accepts wordings, spellings, and capitalizations of answers that have already been entered previously into the quest that is housed on the platform. Answers that are correct but are worded differently will be labeled as "false." Although this allows the platform to constantly update the users on their point status throughout the quest, it also requires all possible wordings and spellings of answers to be devised ahead of time.

### Incorporation of the DE Quest in the Forest Ecology Course

The DE quest was designed as a self-study activity that supports field-based, hands-on laboratory sections of a Forest Ecology course. This course is required by several undergraduate programs in the Faculty of Forestry, UBC, and it focuses on the structure and function of forest ecosystems. A range of topics are covered in this course, including forest productivity, nutrient, carbon and water cycling, soils, the physical environment, population and community ecology, disturbance ecology, ecological succession, biological diversity, and ecological resilience. The overall objective of the course is to provide students with a fundamental understanding of forest ecology as a basis for developing science-based sustainable forest management practices.

The Forest Ecology course is delivered annually from September to December, through campus-based



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1-hour-long lectures three times per week and a total of seven 4-hour-long field-based laboratory sections. The course was established in 1969 and it typically enrolls between 100 and 140 students who are mainly from the Faculty of Forestry but tend to have varying background knowledge of ecology, biology, and soil science.

During the term, students enrolled in the Forest Ecology course will attend a total of seven laboratory sections—two indoor laboratory sections, focused on basic methods of the ecosystem analysis, and five field-based laboratory sections, during which students learn and practice a variety of forest ecosystem description skills. Those skills include: measuring aspect and slope, determining percentage cover by herbaceous and bryophytic species; determining the overstory tree species, cover and dominance class; measuring tree diameter at breast height; describing and classifying soils and the forest floor, and identifying provincial biogeoclimatic and site units. In addition, students also learn how to identify different types of natural and management-induced disturbances within a local (i.e., coastal temperate) forest ecosystem.

Five field-based laboratory sections are carried out in the Pacific Spirit Regional (PSR) Park that is adjacent to the UBC's Point Grey campus (49°15′N, 123°14′W). The Park comprises 763 hectares of forest and foreshore and it includes 73 kilometers of walking trails that allow access to different ecological areas of the park. The park area that Forest Ecology students assess during the field-based laboratory sections is located in the Coastal Western Hemlock very-dry maritime biogeoclimatic zone (Meidinger and Pojar, 1991), dominated with Douglas-fir (*Pseudotsuga menziesii* Mirb.), western hemlock [*Tsuga heterophylla* (Raf.) Sarg.], and western red-cedar (*Thuja plicata* Donn ex D. Don). Unlike many other provincial or regional parks, the PSR Park has experienced a long history of harvesting and fire disturbances. The initial harvesting of the area that is currently included into the PSR Park occurred in the mid-1880s to early 1900s after which the site was allowed to regenerate naturally. There were many fires that impacted the area, some minor and localized and two major in 1886 and 1919 that were initiated by sparks from harvesting operations. The severity of those two major fires varied across the park's area, leading to different types and successional stages of forest that naturally regenerated at different times and areas of the park. Consequently, the PSR Park offers numerous examples of these past management practices and disturbances indicated by a complex pattern of vegetative cover and providing students with rich learning opportunities.

Students are graded based on mid-term and final exams, four laboratory reports, and two laboratory exams. The laboratory exams evaluate student's ability to identify plant species and habitat, including soil nutrient and moisture regimes and types of forest disturbances. Two weeks before the laboratory exams there are no laboratory sections, which allows students to focus on reviewing course content.

The DE quest focused on the following topics: ecosystem succession, ecological disturbances, and disturbance regime, which were chosen because of the rich disturbance history and thus observability in PSR Park. The learning objectives of the DE quest were to (1) enhance students' interest in forest ecology and (2) view the forest ecosystem in an integrative manner. The DE quest was implemented during the 2 weeks before a final laboratory exam. No mark was awarded for the completion of the DE quest; however, students who completed the quest were given the opportunity to have their lowest laboratory report mark taken out of course mark calculation.

## Organizational Structure of the DE Quest

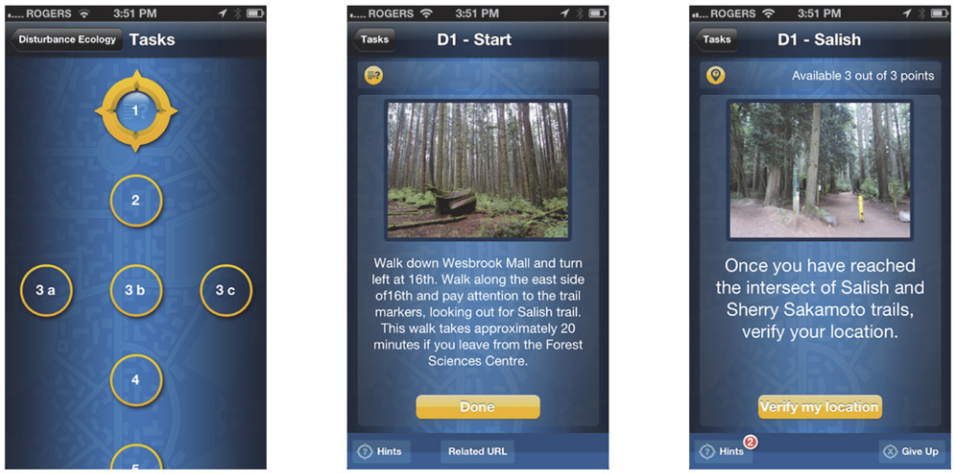
Development of the DE quest included the following key steps: (1) site selection and recording their GPS coordinates, (2) collection of relevant material (e.g., photographs, maps, graphs, online resources), and (3) incorporation of this information into the quest. Site selection involved devising questions based on course learning objectives and making sure that each site within PSR Park provided enough clues to students to answer all questions. The eight sites included in the DE quest were adjacent to sites that students had visited in previous field-based laboratory sections; hence, students were familiar with the PSR Park area before they conducted the self-guided DE quest. The selected sites were illustrative of the following disturbances: forest harvesting (done in the early 1900s), fire (illustrating impacts of two major fires in 1886 and 1919), and windthrow (several events that occurred since the early 1900s). The repetitive visits to the park during DE quest development allowed us to refine the questions and also to determine the best possible order of questions. We made sure that a site photograph accompanied each task that participants encountered on their quest, and more difficult questions were provided with links to websites with relevant information.

Development of the DE quest from initial planning stages and conceptualization to implementation, evaluation, and adjustments was conducted between May and August 2012 and took approximately 150-person-hours. This was much shorter than what development and programming of a web-based educational tool would typically require. In comparison, the development of the web-based tool on Land-Use Impacts on Soil Quality (http://soilweb.landfood.ubc.ca/luitool/) took about 2,000-person-hours (Strivelli et al., 2011). As Polsani (2003) pointed out, development of web-based tools is generally time consuming and requires allocation of a significant amount of time for planning, conceptualization, and discussion as well as other stages of the development process. Mobile apps offer an efficient alternative to development of active learning tools. Development of effective web-based material as well as mobile game-based learning material requires participation of a multidisciplinary team guiding the design (Naidu, 2003; Polsani, 2003); hence, we made sure that diverse expertise was represented on our team. The DE quest was a collaborative effort of students, foresters, scientists, mobile game developers, information technology experts, instructors, and film producers.

The DE quest followed a distinct pattern of instructional tasks, followed by location tasks, and then by a single question or series of questions (Fig. 1 and 2). This ensured that students would be in the right location in the park to answer location-dependent questions. Instructional tasks either directed students to the next location in the quest or provided students with some brief descriptor of where they were (Fig. 1b) and for what they should be looking. Location-based tasks required students to verify their location using the GPS unit on their Apple platform



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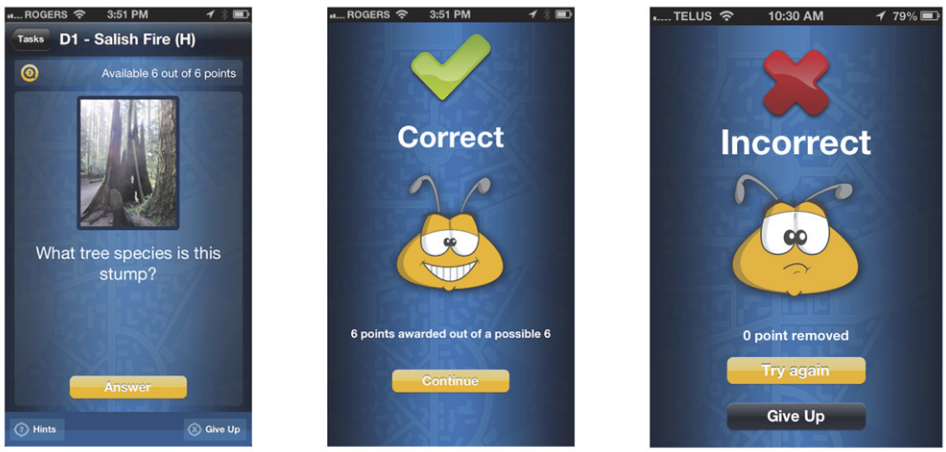

(a) Start screen (b) Initial instructional task (c) Initial location task

Fig. 1. Captures of phone screens showing: (a) start screen, (b) initial instructional task screen, and (c) initial location task of the disturbance ecology (DE) quest.

mobile device (Fig. 1c). The question and answer task (Fig. 2) followed completion of the previous two tasks. Students would be asked either a single question or a series of questions. The questions prompted pre-registered answers as either single-, two- and three-word answers, or multiple-choice.

Out of a total of 16 questions, 9 questions were tiered, with the easiest questions awarded 3 points, moderate questions 6 points, and difficult questions 9 points. Easy questions covered the most basic concepts (e.g., This forest was logged in the 1800s and then burnt in 1910. How old can the oldest trees be then?), while difficult questions focused on topics that were briefly mentioned in the lectures (e.g., What chemical compound is found in the wood of this species that makes it resistant to rot?). Students also received 3 points for correctly verifying their location with the GPS-based tasks. We chose not to deduct any points at any stage, though a built-in function of the Questogo design gives quest designers this option. Scaled questions composed one part of the students' total score on completion of the quest, and the other component was "points balancing." For example, students never had points subtracted for an incorrect score, but they could "point-balance" by cashing in 1 to 3 points for hints to questions that they did not know the answer to. Point balancing ensures that students were not being disadvantaged for answering a question incorrectly on their first try. This was especially important in the context of the DE quest because occasionally students would come up with the right answer, but would have to try out a few different wordings of the answer before the app would accept it as correct. Upon submission of an incorrect answer, students would be shown a screen telling them that they were incorrect, and that they should try again (Fig. 2c). Students did not have points deducted for writing in an incorrect answer, and were provided opportunities to submit another answer if they were registered as incorrect. Overall, the different number of points awarded intended to give the DE quest a competitive notion and thus increase the gaming-feel of the exercises.

Students completed the DE quest in about 2 hours and worked in groups of four to six students. This promoted




(a) Question and answer task (b) Correct answer for the question and answer task (c) Incorrect answer for the question and answer task

Fig. 2. Captures of phone screens showing screens for: (a) question and answer task, (b) correct answer, and (c) incorrect answer for a question task of the disturbance ecology (DE) quest.



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Table 1. Online survey questions regarding the Disturbance Ecology (DE) quest given to the Forest Ecology students.

| | Question |
|---|---|
| 1 | What type of information would you have liked to have before heading out on this quest? |
| 2 | How would you evaluate the level of difficulty of the tasks and questions presented in this quest? |
| 3 | What would you say you learned from participating in this quest? |
| 4 | Were the questions and tasks presented with sufficient clarity? Specifically, were there any questions that were confusing? |
| 5 | Did you find this quest easy to use, and would you recommend it to be used in the future? |
| 6 | What other questions would you like to see in this type of quest? (Please provide additional comments) |
| 7 | Did you find the Questogo App easy to use? If not, what parts did you have problems with? |
| 8 | Was there anything you did not like about the app? Anything you liked? (Please provide additional comments) |
| 9 | Now that you have completed the Disturbance Ecology (DE) quest, are you interested in doing/planning to do other quests? |

team work and allowed students to have face-to-face interaction with their peers, while still giving them the liberty of completing the quest on their own if they chose to. Since the DE quest was based on the Questogo platform, which only supports Apple mobile devices, students shared iPhones that were provided by the developer company (14Oranges Software Inc., Richmond, BC) or used their own mobile devices. The Questogo platform did not have the social media sharing feature at the time of use, since Questogo was still in a pilot stage; hence, students were not able to log onto social media sites such as Facebook and Twitter to view results of other groups participating in the DE quest. After completion of the quest, students were asked to submit a group written report that needed to include the group's total number of points achieved and photos of the group members participating in the quest.

## Feedback Collection

The DE quest was implemented in the Forest Ecology course during the 2011–2012 academic year and the 130 students enrolled in the course were asked to complete an online feedback form. The response rate was 40% ($n$ = 52). The form was modeled after design-based research principles (Barab and Squire, 2004; Wang and Hannafin, 2005), which provided participants with a complete disclosure of the survey intentions (i.e., quest description, its learning outcomes, and the overall study objective). The feedback form included nine open-ended questions (Table 1); responses were given in a multiple-choice format, with room for comments after each question. As an incentive for students to complete the survey we offered the chance to win a gift card to a local coffee shop.

## RESULTS

For all survey questions, students were allowed to select more than one answer, resulting in totals of more than 100% for most responses. Students were not provided background information on Questogo before initiating the DE quest and were not told the exact park location where they were supposed to finish the quest. Students were provided treasure-hunt style directions to the starting location of the DE quest. It was assumed that all students would be starting from the faculty of the Forestry building, which is a 20-minute walk away from the PSR Park where DE quest was carried out. To learn if students felt as though they were provided with sufficient information to adequately complete the DE quest, we included the following question into the survey: What type of information would you have liked to have before heading out on this quest? Forty-one percent of all students who completed the online survey stated that that there was enough information provided to complete the quest, whereas 16% thought there could be more location tasks provided, and another 19% indicated that there were too many location tasks (Fig. 3a).

In response to the question, How would you evaluate the level of difficulty of the tasks and questions presented in this quest? (Fig. 3b), 88% of students thought that the DE quest was designed at an appropriate level of difficulty, and that it incorporated suitable types of material for review. Six percent of students indicated that questions were too difficult and another 6% thought that questions were too easy. When asked, What would you say you learned from participating in this quest? (Table 2), 34% of students indicated that they did not learn anything new from the DE quest, but they still found the quest to be a good review of the course material. Another 47% of respondents indicated that the DE quest helped them deepen their knowledge of material covered in the Forest Ecology course, and 44% of students indicated that the quest allowed them to learn more about the topic of disturbance ecology (Table 2).

When asked, Were the questions and tasks presented with sufficient clarity? Specifically, were there any questions that were confusing? (Table 2), 16% of students thought that questions were clear and 81% stated that questions were clear with the occasional exception. In the comments following this question, students identified questions that were found to be unclear. These were mainly the questions that required short phrases for answers, instead of single-word answers. The Questogo platform will only accept pre-programmed answers; hence, answers that were one letter off or were spelled incorrectly were not recognized as correct. Although multiple answers were pre-programmed into the DE quest, some options of a correct answer were still not included. From responses and suggestions received it became clear that questions should either have one-word answers or we would need to specify how many words are required in the answer.

Despite the issue of having a few confusing questions, a majority of the respondents (81%) stated that they would recommend the DE quest to other students when asked, Did you find this quest friendly to use, and would you recommend it to be used into the future? (Table 2). In addition, 19% of students stated that they would recommend the DE quest for younger (e.g., high school) students, whereas 13% would recommend it for older students. Three percent of respondents indicated that they would not recommend this quest to others.

Since this quest was designed as a review tool for students preparing for their final laboratory exam, we felt that it was important to find out whether students thought that the questions covered the right type of material. Students were asked, What other questions would you



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Table 2. Online survey student responses ($n$ = 52) to the questions 3, 4, 5, and 6 regarding the the Disturbance Ecology (DE) quest.

| Answer choices | Responses (%)† |
|---|---|
| Question 3: What would you say you learned from participating in this quest? | |
| I did not learn anything new, but the quest provided a good review of course material | 34 |
| I learned material covered in the Forest Ecology course | 47 |
| I learned more about disturbance ecology | 44 |
| I did not learn anything | 0 |
| Question 4: Were the questions and tasks presented with sufficient clarity? Specifically, were there any questions that were confusing? | |
| Questions were clear | 16 |
| Questions were clear with the occasional exception | 81 |
| Questions were not clear and were confusing | 6 |
| Question 5: Did you find this quest easy to use, and would you recommend it to be used into the future? | |
| Yes, I would recommend this quest to other students | 81 |
| No, I would not recommend this quest to other students | 3 |
| Yes, but I would recommend this quest for younger students (such as high school groups) | 19 |
| Yes, but I would recommend this quest for older students | 13 |
| Question 6: What other questions would you like to see in this type of quest? | |
| More questions focusing on plant identification | 59 |
| Less location task questions | 25 |
| More location task questions | 13 |
| More questions on forest ecology | 50 |
| More questions focused on ecology | 41 |

† Responses add up to more than 100% because students were allowed to choose multiple answers for the question.

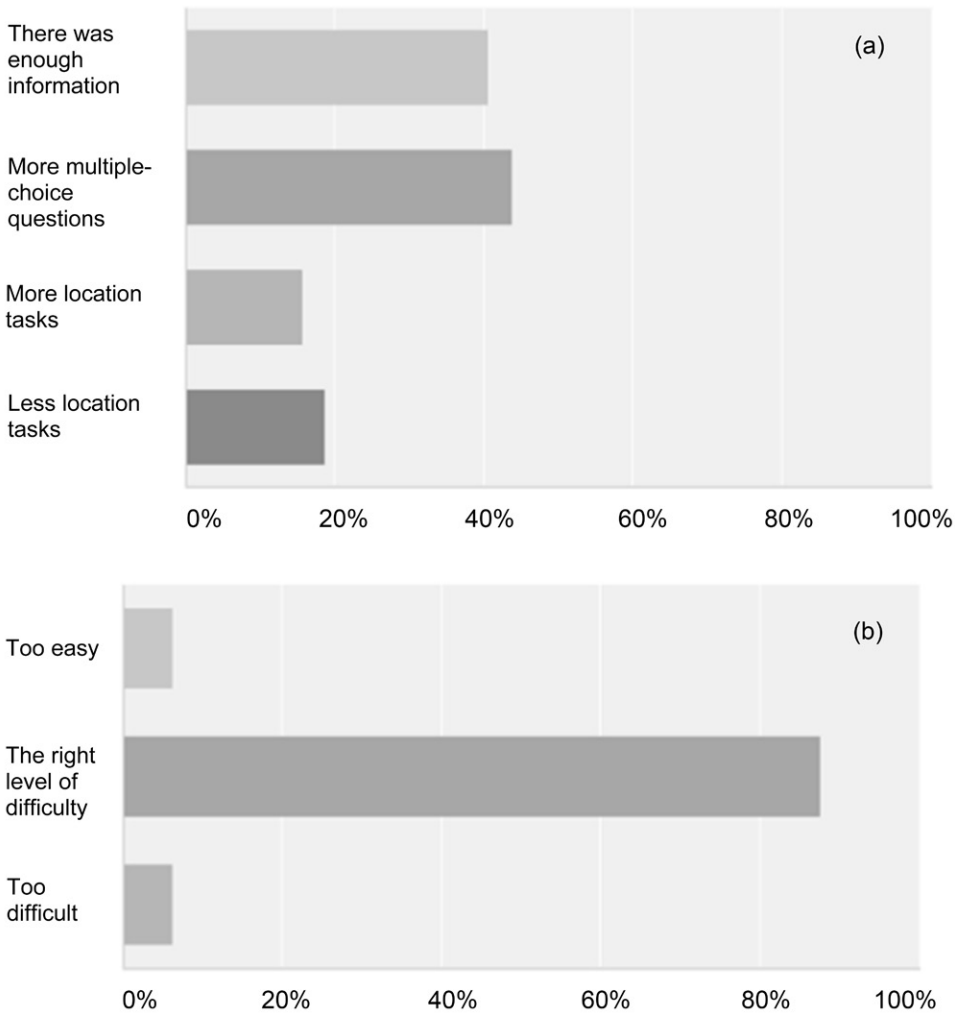



Fig. 3. Online survey student responses ($n$ = 52) to the questions: (a) What type of information would you have liked to have before heading out on this quest? and (b) How would you evaluate the level of difficulty of the tasks and questions presented in this quest?



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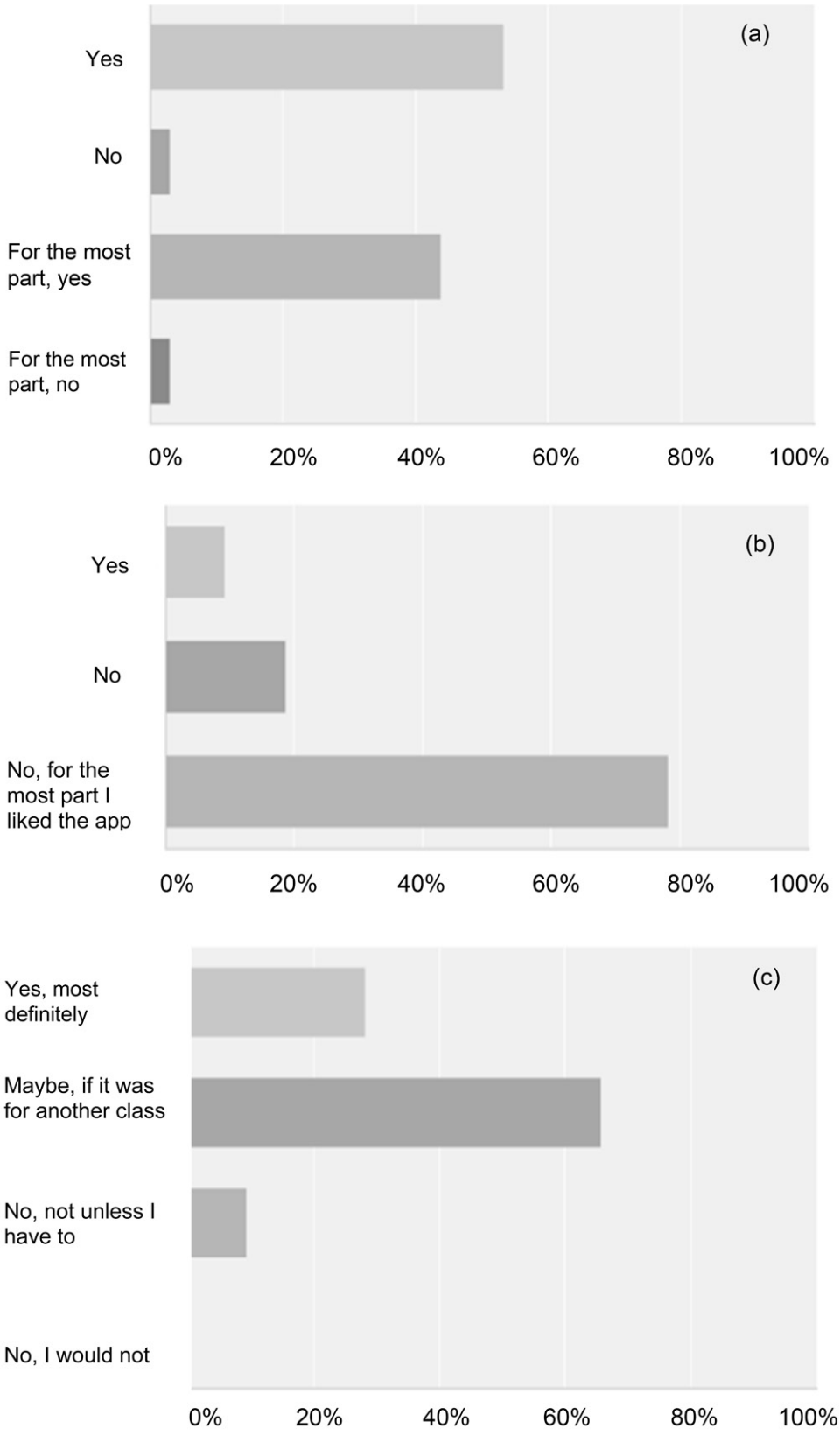



Fig. 4. Online survey student responses ($n$ = 52) to the questions: (a) Was there anything you did not like about the Questogo App? (b) Was there anything you did not like about the app? Anything you liked? Please specify in comment section, and (c) Now that you have finished the DE quest, are you interested in doing/planning to do other quests?



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like to see in this type of quest? and were requested to provide ideas as comments (Table 2). Since one of the two laboratory exams focused on plant identification, it was not surprising to find out that 59% of students indicated that they would like to have a higher number of plant identification questions (Table 2). Students also indicated that they wished to see a higher total number of questions on the broader forest ecology topics and on ecology in general, with 50 and 41% indicating those preferences, respectively.

The ease of use of new technology is critical in ensuring that students are actually able to engage with the content without having the technology present any barriers. The majority of students (96%) stated that they found the app easy to use or easy to use in most situations when asked, Did you find the Questogo App easy to use? If not, what parts did you have problems with? (Fig. 4a). Three percent of students indicated that they did not find the quest easy to use, and an additional 3% indicated that they did not find the quest easy to use in most situations (Fig. 4a).

When asked, Was there anything you did not like about the app? Anything you liked? Please comment, 78% of students stated that "for the most part I liked the app" (Fig. 4b). In addition, 19% indicated that they did not have any issues with the app, whereas 9% of students stated that they did not like the app. Through students' comments it became clear that wording of answers (as detailed above) was the main issue of dissatisfaction with the app. When students answered correctly, but the app did not accept the wording of their answer, students became frustrated, with one student commenting, "Typing in answers was difficult because our answers had to exactly match your answers and sometimes the wording was tough to guess." Another student looked more favorably on the DE quest, despite running into issues, stating that, "I liked not being deducted points for wrong answers, and the questions were about the environment around us, it was a self-guided lab."

When asked, Now that you have completed the DE quest, are you interested in doing/planning to do other quests? (Fig. 4c), 28% of students stated that they would most definitely be interested in doing other quests in the Forest Ecology course, and 66% indicated that they would be interested in doing quests if they were incorporated into other courses. Another 9% indicated that they would only do a quest again if they had to, and no students stated that they would never do a quest again.

## DISCUSSION

There is an on-going need for forestry and land resource professionals to gain a solid understanding of how to sustainably manage our forests. One approach for acquiring this knowledge is to enhance current educational methods with innovative, active learning approaches. Mobile devices—with their portability, social interactivity (as both face-to-face interaction and exchange of data between learners), connectivity, and flexibility to support different learning activities—are well suited to foster learning outside traditional educational settings (Klopfer and Squire, 2008). The DE quest is a mobile game-based learning resource, developed to help meet the challenge of educating the current "net generation" of students as future forestry and land resource professionals. The DE quest was developed not as a replacement for the outdoor laboratory sections but as a resource that supports those sections of the Forest Ecology course. Implementation of the DE quest as a self-study activity that supplements traditional laboratory sections is in agreement with other studies that found mobile game-based learning to be well suited to support traditional classes or e-Learning courses (Tsai et al., 2012). This technology has potential in other learning environments as well, such as distance education, field-based workshops, or indoor laboratories.

Student responses to the DE quest were generally positive. Students' feedback showed that the DE quest was designed at an appropriate level to help them carry out a self-study review of material covered in the laboratory sections. Of the feedback gathered, some common responses arose indicating certain shortcomings in the design of the content for the Questogo App. The most common comment was that the app needed to accept more wordings for the written answer questions. Students suggested that having more multiple-choice questions would remedy this situation. But the feedback also showed strengths of the Questogo App because students found the user interface as well as the navigation of the app to be user-friendly. Furthermore, the majority of students found the DE quest to be a useful self-study tool to support learning of concepts covered in the outdoor laboratory sections of the Forest Ecology course. A major benefit of the DE quest is that it allows students to go back to the app and review the questions asked and topics covered as many times as they wish. In a typical field-based laboratory section of the Forest Ecology course, a group of up to 60 students visited various sites with one instructor, limiting direct hands-on participation time by each student. The DE quest and the app allowed students to learn concepts that they may have missed during the outdoor laboratories because they were said out of earshot. This was in agreement with findings of other studies that online and mobile educational tools reinforce learning by allowing review of content as many times as needed (Polsani, 2003), helping the development of deeper knowledge (Najjar, 1995).

Most students (i.e., 81% of respondents) indicated that they were able to engage with the mobile game-based leaning technology and found the experience enjoyable. The majority of students commented that they would like to see quests incorporated into other courses (66%) as well as having additional quests in the Forest Ecology course (28%). This was very encouraging, especially since students did encounter some technical issues with the DE quest, namely regarding written answer questions that needed to be pre-programmed in the app.

The student responses indicated that mobile game-based learning was effective as a supplement to field-based and in-class instruction and highlighted promising new directions for future implementations of mobile technologies in education.

## ACKNOWLEDGMENTS

We appreciate the assistance and insights of Mr. Sylvain Marcotte (14Oranges Software Inc., Richmond, BC) and Mr. J. Joly (Overinteractive Media, Vancouver, BC) during DE quest development. We would also like to thank Dr. Les Lavkulich (Faculty of Land and Food Systems, UBC) and Dr. Jolie Mayer-Smith (Faculty of Education, UBC) for their help with this study. This project was carried out as a part of the



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Terrestrial Research on Ecosystems & World-wide Education & Broadcast (TerreWEB) graduate training program at UBC, funded by the Natural Sciences and Engineering Research Council (NSERC) of Canada Collaborative Research and Training Experience (CREATE) initiative.

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